

Transcript

Slide 1

Good afternoon. My name is Anne. One of the pressing issues throughout this module, and also an issue I face within my own profession, is: How do organisations manage AI innovation with regard to security and ethics? I used to see governance as paperwork: just document it, submit it, move on. However, I have since reconsidered this perspective. Governance is more than just paperwork. It is essential to the trustworthiness of AI within a live environment, among live users, and within decision-making that impacts human life. My view now is that monitoring matters more than documentation, and contestability connects the two.

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Two cases made this clear to me. First is Amazon's recruitment programme, built on ten years of hiring data. It ended up downgrading female candidates because it copied patterns in the data. As Dastin (2018) reported, "a decade of data reflected the imbalance of women in technology jobs." There was nothing prejudicial in the system's design; bias was simply inherited through training data, as the saying goes: "bias in, bias out." Secondly, the SyRI system in the Netherlands was more likely to scrutinise individuals from lower-income areas for suspected welfare fraud. In 2020, the District Court of The Hague ruled that SyRI was in violation of human rights. Together, these cases raise a simple question: Is technical correctness sufficient for a system to be ethical?

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Regulators have recognised this too. There are three regulatory frameworks now in place: the EU AI Act (European Union, 2024), the NIST Risk Management Framework (NIST, 2023), and the OECD Principles on Artificial Intelligence (OECD, 2024). Each of these frameworks has emphasised key principles: transparency, accountability, and human oversight. In 2024, Sousa e Silva discussed how the EU AI Act imposes binding legal obligations for high-risk systems. But none of these frameworks sufficiently deals with what happens when a system behaves unpredictably after deployment. That gap is what governance scholars call a socio-technical gap.

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This gap creates security vulnerabilities. Three major types of attack exploit it: input attacks that fool models, poisoning attacks that undermine the learning process, and intellectual property theft that targets models. Salem and colleagues (2024) categorise these attacks, while Li and Zhang (2025) confirm how they are already impacting organisations in practice. But countermeasures do exist: differential privacy, adversarial training, federated learning. These, however, are technical controls that mitigate threats, rather than a complete cure for governance risks. Beyond these technical threats, deepfakes pose unique challenges. Research by Fu, Hadid, and Damer (2025) explores detection difficulties. A good example is Arup in 2024. Chan (2024) reported how fraudsters deepfaked senior executives on a video

call and convinced an employee to release 25 million dollars without breaching any system. Visual trust is now vulnerable. Arup was not an isolated case. Deloitte (2024) projects that AI-enabled fraud could reach 40 billion dollars by 2027.

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The underlying problem is that most governance approaches assume AI systems stay stable. But they don't stay stable. They shift as they learn from live data, so they change after deployment. Tamang (2025) calls this distribution shift. He explains that the problem isn't faulty models; it's the mismatch between training conditions and real-world operation. Drift needs to be constantly tracked in real-time, but there's no reliable method to do that. For businesses, this means legal exposure, remediation costs, and delayed adoption.

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The healthcare sector is the most illustrative example in this regard. AI systems can identify promising drug candidates in weeks rather than years. The benefit is clear. Here's the paradox, though: the more a system is governed, the more it is adopted. Why? Regulators trust documented systems. Clinicians trust transparent tools. Microsoft, with its RAI Standard, and Google, with its Model Cards, understood this early. There is a tension here. Governance demands transparency, but the most accurate AI models are often the hardest to explain. Nouis, Uren and Jariwala (2025) call this a deeper conflict in healthcare AI.

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So, can this tension be resolved in practice? I believe so. To validate my argument, I designed a prototype: a retrieval-and-generation system with contestability embedded. The aim is to show how governance can work in practice, not just on paper. In this prototype, three mechanisms are used. Firstly, drift alerts trigger when outputs deviate more than 2% from baseline, proportional thresholds, not absolute. Secondly, confidence below 85% requires human review, matching the EU AI Act's human-on-the-loop principle (European Union, 2024). Thirdly, decision logs create full audit trails. Time to detection, time to escalation, and overturn rates are calculated, and the response to escalation is to be within 4 hours. This works in a prototype. But is it realistic at scale?

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A common criticism is that governance slows innovation, especially in small businesses. Compliance costs money. That is true: Costs rise with the level of governance, human oversight takes time, and paperwork is expensive. But governance is becoming a competitive advantage, and this year, complying with the EU AI Act is going to be the new norm in enterprise procurement. Saeri and colleagues (2025) identify formal governance and oversight mechanisms as risk mitigations intended to reduce harmful AI incidents.

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So what ties this together? Three components, one organising principle. Firstly, adaptive structures that focus on continuous monitoring, not intermittent compliance, as emphasised by Scaramuzza and colleagues (2025). Secondly, security that is multi-layered, combining technical controls with human validation. Thirdly, accountability that is driven by human intervention at critical decision points. The glue that holds all these elements together is contestability, which is the ability to challenge, review, and override. Ethical argumentation, regulatory analysis, and the functional prototype all point to the same conclusion: AI governance should be operational, not bureaucratic.

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When I started this module, I saw governance as a brake on innovation. Document it, submit it, move on. I was wrong. Amazon showed me the cost of inherited bias. SyRI showed me the price of unchallenged automation. My prototype showed that governance can be incorporated from the start. So how do we balance innovation with ethics and security? We don't choose between them. When contestability is built in, the two become the same thing. That's not a burden. That's the point.

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